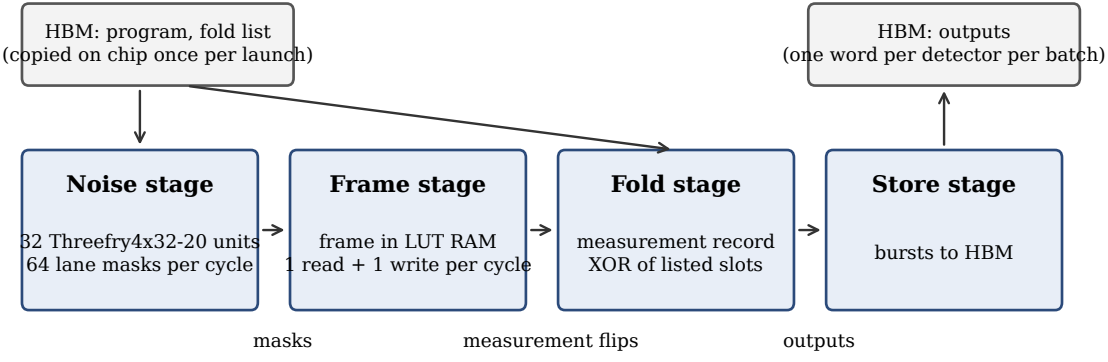
HBM: program, fold list
(copied on chip once per launch)
HBM: outputs
(one word per detector per batch)
Noise stage
32 Threefry4x32-20 units
64 lane masks per cycle
Frame stage
frame in LUT RAM
1 read + 1 write per cycle
Fold stage
measurement record
XOR of listed slots
Store stage
bursts to HBM
masks
measurement flips
outputs
all loops pipelined at one iteration per cycle; stages connected by FIFO streams